

Matrix inequalities and norms

The Marcus–de Oliveira determinantal conjecture

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Rating rationale: This decades-old spectral enclosure conjecture for arbitrary normal pairs is a major matrix-theory barrier with consequences for determinant and spectral analysis.

Problem statement

For every integer $n \geq 1$, let $A, B \in \mathbb{C}^{n \times n}$ be normal matrices, meaning $AA^* = A^*A$ and $BB^* = B^*B$. List their eigenvalues with algebraic multiplicity as $a_1, \dots, a_n$ and $b_1, \dots, b_n$. Is

$$\det(A + B) \in \operatorname{conv} \left\{ \prod_{i=1}^n (a_i + b_{\sigma(i)}) : \sigma \in S_n \right\}?$$

Here S_n is the permutation group and the convex hull is taken in $\mathbb{C} \cong \mathbb{R}^2$.

Why it matters

The conjecture would bound a determinant using only two spectra, even when the matrices are not simultaneously diagonalizable. It is a fundamental spectral enclosure problem for structured matrix computations.

References

1. A. Kovacec, *A conjecture more precise and stronger than the one by Marcus and de Oliveira*, DMUC Preprint 26-11 (7 April 2026), §1, Conjectures 1 and 1'. [Primary paper](#).
2. N. Bebiano and J. P. da Providência, *Revisiting the Marcus–de Oliveira conjecture*, *Mathematics* 13(5) (2025), 711, §1. [DOI](#).
3. J. I. Mulero-Martínez, *A variational framework for determinantal inequalities of normal matrices: Successes and obstructions*, *Linear Algebra and its Applications* 740 (2026), 19–38; abstract and structured-class results. [DOI](#).

Status check — 2026-09-10

The April 2026 preprint proposes a stronger conjecture rather than proving this one. The July 2026 journal paper explicitly describes the unrestricted conjecture as open and proves structured cases. Searches included *Marcus de Oliveira conjecture 2026 proof*, *Marcus Oliveira determinant counterexample*, and the exact 2026 titles. No complete proof or counterexample was located. Primary full text was checked for reference 1; reference 3's status evidence was its publisher abstract, not an independently audited proof.

Audit update (2026-09-10): Rechecked Kovacec's Conjecture 1 and the July 2026 publisher abstract of Mulero-Martínez, then searched for a full resolution. The latter explicitly retains the general conjecture while proving structured spectral classes. This is a bounded literature check, not a proof that no solution exists.